

# Alexander Sierra

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## SUMMARY

Software developer with experience building machine learning pipelines, webapps, and data-driven applications with Python and Rust.

## SKILLS

**Languages:** Python, Rust, TypeScript, R

**Web:** HTML, Leptos, Axum, NextJS, Tailwindcss

**Tools:** Git, VS Code, Jira, OpenAI API, Microsoft Office Suite

## PROJECTS

### curatedTBExplorer — Data Exploration Web App (R, Shiny)

- Built an interactive application for analyzing tuberculosis gene expression data and using ensemble learning techniques to explore gene signatures that distinguish patients with PTB, LTBI, and control.
- Developed filtering pipeline to handle missing values and ensure dataset integrity, as well as implementing options to subset training data based on patient features, enabling targeted gene signature exploration.

### cal-detect — Computer Vision TCN Training Pipeline (Python, PyTorch)

- Designed and implemented a training pipeline for a Temporal Convolutional Network (TCN) to classify the validity of individual calisthenics repetitions from MediaPipe Landmark features.
- Built data preprocessing logic to normalize landmarks and derive velocity and angle features.

### quic-analysis — QUIC Traffic Analysis & Classification (Python, Selenium, tshark)

- Developed a pipeline to simulate and capture QUIC web traffic using Selenium WebDriver and tshark.
- Extracted traffic flow features including packet size, directionality, and interarrival time from recorded traffic.
- Trained a convolutional neural network for classifying websites based on extracted traffic flow features.

### mock-interview-agentic-system — User Interactive Agentic System (Python, OpenAI API, AutoGen, TTS, STT)

- Implemented an Agentic System Pipeline using AutoGen, to dynamically interview the user with a resume and job description as context.
- Used Vosk and Piper to implement text-to-speech and speech-to-text capabilities.

## EXPERIENCE

### IT Helpdesk Technician | Southern Utah University, Cedar City UT

May 2023 – Sep 2024

- Resolved and documented support tickets involving authentication, networking, and system issues.
- Provided troubleshooting support using remote access tools across Windows/macOS environments.
- Documented incidents and resolutions in Jira.
- Registered computers in the network DHCP table.

### Phone Operator | AvantGuard, Cedar City UT

Jun 2023 – Oct 2023

- Promptly responded to alarm notifications and reported incidents to emergency dispatch.

## EDUCATION

### MS in Computer Science | University of Colorado Denver, Denver CO | GPA 3.8

Jan 2025 – May 2026

**Relevant Courses:** GenAI, Deep Learning, Algorithms, Cybersecurity Programming, Computer Vision

### BS in Computer Science | Southern Utah University, Cedar City, UT | GPA 3.7

Aug 2020 – Apr 2024

**Relevant Courses:** Operating Systems, Data Comm. & Networking, Database Design & Management